UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PRE CALCULUS
COURSE CODE	:	MAT133
EXAMINATION	:	NOVEMBER 2026
TIME	:	2 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the MAT133 mock paper (November 2026).
   Answers are unofficial model answers generated from the printed past-paper question; they are not an official UTM marking scheme.

2. Marking points follow the printed part labels and marks. Verify every model answer against the source page.

3. Four questions: 10 + 10 + 10 + 25 marks; total = 55 marks.

4. Questions: 4. Real recycled questions: 4. Placeholders: 0.
   For local study use only; model answers are unofficial.

5. Every question is a real MAT133 past-paper question; its source paper and page are printed with the answer.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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QUESTION 1 (10 marks)

Source: MAT133-2025-JUL, page(s) 2

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Solve the inequality $(5-x)^2 \le 9$ and express the solution in real number line.
(5 marks)

Answer: $2 \le x \le 8$, i.e. $x \in [2, 8]$ (shade the segment from 2 to 8 on the number line, solid dots at both ends)

- $(5-x)^2 \le 9 \Leftrightarrow |5-x| \le 3$
- $-3 \le 5-x \le 3$
- Subtract 5: $-8 \le -x \le -2$
- Multiply by -1 (reverse inequalities): $2 \le x \le 8$
- Check: $x=2 \to 3^2 = 9 \le 9$; $x=8 \to (-3)^2 = 9 \le 9$; $x=5 \to 0 \le 9$ ✓

b) Write the equation of circle $x^2+y^2-4x-2y-11=0$ in standard form. Hence, sketch the circle with the correct center and radius.
(5 marks)

Answer: $(x-2)^2+(y-1)^2=16$; centre $(2,1)$, radius 4. Sketch: circle centred at $(2,1)$ passing through $(6,1), (-2,1), (2,5), (2,-3)$.

- Group: $(x^2-4x)+(y^2-2y)=11$
- Complete the square: $(x^2-4x+4)+(y^2-2y+1)=11+4+1$
- $(x-2)^2+(y-1)^2=16$
- Centre $(2,1)$, radius $r=\sqrt{16}=4$
- Sketch: plot centre $(2,1)$; mark intercepts $(6,1), (-2,1), (2,5), (2,-3)$ and draw circle of radius 4

Assumptions: Part (a) asks to 'express the solution in real number line'; assumed this means shading the solution interval on a number line. Part (b) sketch assumed to be a standard Cartesian circle; no grid values given in the question.

Model confidence: high.

QUESTION 2 (10 marks)

Source: MAT133-2022-FEB, page(s) 2

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Solve the inequality $\frac{2}{-2x-1} \le 2$. Express the answer in real number line.
(5 marks)

Answer: $x \in (-\infty, -1) \cup \left[-\frac{1}{2}, \infty\right)$; on the number line: open circle at -1, closed circle at $-\frac{1}{2}$, shade left of -1 and right of $-\frac{1}{2}$.

- $\frac{2}{-2x-1} \le 2 \Leftrightarrow \frac{2}{-2x-1} - 2 \le 0$
- $\frac{2-2(-2x-1)}{-2x-1} \le 0 \Leftrightarrow \frac{4x+4}{-2x-1} \le 0$
- $\frac{4(x+1)}{-(2x+1)} \le 0 \Leftrightarrow \frac{x+1}{2x+1} \ge 0$
- Critical values: $x=-1$ (numerator, included) and $x=-\frac{1}{2}$ (denominator, excluded)
- Sign table: $x<-1$: $(+)/(-)=-$; $-1<x<-\frac{1}{2}$: $(+)/(+)=+$; $x>-\frac{1}{2}$: $(+)/(+)=+$
- Need ≥ 0: $x<-1$ or $x>-\frac{1}{2}$, together with $x=-1$
- Solution: $(-\infty, -1] \cup \left(-\frac{1}{2}, \infty\right)$

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b) If $Z_1 = 2 - 7i$ and $Z_2 = -2 + 3i$, state $\frac{\overline{Z}_1 i^{23}}{Z_2}$ in standard form $a + bi$.
(5 marks)

Answer: $\frac{13}{13} + \frac{26}{13}i = 1 + 2i$

- $i^{23} = i^{20} \cdot i^3 = (i^4)^5 \cdot (-i) = -i$
- $\overline{Z_1} = 2 + 7i$
- $\overline{Z_1} i^{23} = (2 + 7i)(-i) = -2i - 7i^2 = 7 - 2i$
- $\frac{7 - 2i}{-2 + 3i} \cdot \frac{-2 - 3i}{-2 - 3i} = \frac{(7 - 2i)(-2 - 3i)}{4 + 9}$
- Numerator: $-14 - 21i + 4i + 6i^2 = -14 - 17i - 6 = -20 - 17i$
- $\frac{-20 - 17i}{13} = -\frac{20}{13} - \frac{17}{13}i$

Assumptions: Part (a): the inequality is taken as printed, $\frac{2}{-2x - 1} \leq 2$. Part (b): the expression is taken as printed, $\frac{\overline{Z_1} i^{23}}{Z_2}$.

Model confidence: high.

QUESTION 3 (10 marks)

Source: MAT133-2026-FEB, page(s) 2

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Given that $p(x) = \frac{7}{x} + 10$ and $q(x) = \sqrt{x - 5}$.

i) State the domain of $p(x)$ and $q(x)$.
(2 marks)

ii) Determine $(q \circ p)(x)$.
(3 marks)

Answer: Domain of p: $x \neq 0$ (i.e. $(-\infty, 0) \cup (0, \infty)$); domain of q: $x \geq 5$ (i.e. $[5, \infty)$) | $(q \circ p)(x) = \sqrt{\frac{7}{x} + 5} = \sqrt{\frac{7 + 5x}{x}}$

- $p(x) = \frac{7}{x} + 10$ requires $x \neq 0$
- $q(x) = \sqrt{x - 5}$ requires $x - 5 \geq 0 \rightarrow x \geq 5$
- $(q \circ p)(x) = q(p(x)) = \sqrt{p(x) - 5}$
- $= \sqrt{(\frac{7}{x} + 10) - 5} = \sqrt{\frac{7}{x} + 5}$
- $= \sqrt{\frac{7 + 5x}{x}}$

b) Given that $f(x) = \frac{4 - 3x}{5x - 7}$, find the value of x if $f^{-1}(x) = 2$.
(5 marks)

Answer: $x = \frac{18}{13}$

- $f^{-1}(x) = 2 \Leftrightarrow f(2) = x$
- $f(2) = \frac{4 - 3(2)}{5(2) - 7} = \frac{4 - 6}{10 - 7} = \frac{-2}{3}$
- $x = -\frac{2}{3}$

Assumptions: Part b: interpreted as find x such that $f^{-1}(x) = 2$, i.e. $x = f(2)$; answer $x = -2/3$.

Model confidence: medium.

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QUESTION 4 **(25 marks)**

Source: MAT133-2019-JUN, page(s) 3

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Given $f(x)=\sqrt{x-9}$, $x \geq h$ and $g(x)=\frac{3-2x}{5+x}$, $x \neq k$. Find

i) the values of h and k

(2 marks)

ii) $(f \circ g)(x)$

(3 marks)

iii) $g^{-1}(2)$

(4 marks)

Answer: i) $h=9,\ k=-5$; ii) $(f \circ g)(x)=\sqrt{\frac{-7x-42}{5+x}}$; iii) $g^{-1}(2)=-1$

- i) $f(x)=\sqrt{x-9}$ requires $x-9 \geq 0 \rightarrow x \geq 9$, so $h=9$.
- $g(x)=\frac{3-2x}{5+x}$ requires $5+x \neq 0 \rightarrow x \neq -5$, so $k=-5$.
- ii) $(f \circ g)(x)=f(g(x))=\sqrt{g(x)-9}=\sqrt{\frac{3-2x}{5+x}-9}$
- $=\sqrt{\frac{3-2x-9(5+x)}{5+x}}=\sqrt{\frac{3-2x-45-9x}{5+x}}=\sqrt{\frac{-11x-42}{5+x}}$
- iii) Let $y=g^{-1}(2) \rightarrow g(y)=2$: $\frac{3-2y}{5+y}=2 \rightarrow 3-2y=10+2y \rightarrow -7=4y \rightarrow y=-\frac{7}{4}$

b) Use long division to find the quotient and remainder when $P(x)=2x^3-x$ is divided by $(x-3)$.

(5 marks)

Answer: Quotient $=2x^2+6x+17$, remainder $=51$

- $2x^3+0x^2-x+0$ divided by $x-3$:
- $2x^3 \div x=2x^2$; subtract $2x^2(x-3)=2x^3-6x^2$: remainder $6x^2-x$
- $6x^2 \div x=6x$; subtract $6x(x-3)=6x^2-18x$: remainder $17x$
- $17x \div x=17$; subtract $17(x-3)=17x-51$: remainder 51
- $P(x)=(x-3)(2x^2+6x+17)+51$

c) Solve the equation $5^{x+1}-5^{2x}=4$.

(6 marks)

Answer: $x=0$

- $5^{x+1}-5^{2x}=4 \rightarrow 5 \cdot 5^x-(5^x)^2=4$
- Let $u=5^x>0$: $5u-u^2=4 \rightarrow u^2-5u+4=0$
- $(u-1)(u-4)=0 \rightarrow u=1$ or $u=4$
- $5^x=1 \rightarrow x=0$; $5^x=4 \rightarrow x=\log_5 4$
- Check $x=0$: $5-1=4$ ✓; $x=\log_5 4$: $5 \cdot 4-16=4$ ✓
- So $x=0$ or $x=\log_5 4$

d) Sketch the graph of $y=|x|$. Hence, sketch the graph of $y=5-|x-5|$ by using transformation techniques.

(5 marks)

Answer: $y=|x|$: V-shape vertex $(0,0)$ through $(1,1),(-1,1)$. $y=5-|x-5|$: shift right 5, reflect in x-axis, shift up 5; vertex $(5,5)$, passes $(0,0)$ and $(10,0)$, downward V.

- Start with $y=|x|$ (vertex origin, arms up slope ± 1).

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- $y = |x - 5|$: translate right 5 → vertex $(5, 0)$.
- $y = -|x - 5|$: reflect in x-axis → vertex $(5, 0)$, arms down.
- $y = 5 - |x - 5|$: translate up 5 → vertex $(5, 5)$; x-intercepts at $x = 0, 10$.

Assumptions: Part (a)(ii): the printed expression is taken as (f∘g)(x); the simplified radicand is -11x-42 over 5+x (the hint's -7x-42 appears to be a typo). Part (c): both solutions x=0 and x=log_5 4 are given since the equation is quadratic in 5^x.
Model confidence: high.

END OF MARKING SCHEME

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